

MooreVIEW

APG True Echo Radar — Device Review

Modbus RTU level transmitter integration

Template: `apg_true_echo_rtu`

Document generated: 2026-08-12

Integrator review — register map, appliance tags, Opta bring-up

MooreVIEW — APG True Echo Radar Device Review

Vendor: Automation Products Group (APG) — apgsensors.com
Models: True Echo CR-L / True Echo Plus (RS-485 Modbus RTU)
MooreVIEW template: `apg_true_echo_rtu`
Document purpose: Integrator / engineering review of Modbus map, appliance tags, and Opta bring-up
Generated: 2026-08-12

1. Summary

MooreVIEW adds first-class support for APG **True Echo** radar level transmitters as a dedicated Modbus RTU instrument:

Path	Role
PC appliance	Hardware Wizard → Modbus RTU → APG True Echo (<code>src/devices/templates/apg_true_echo_rtu.json</code>)
Opta sample	Standalone RS-485 poll sketch (<code>firmware/arduino-opta-true-echo/</code>)
Probe	<code>node scripts/modbus-probe.js --serial COMx --baud 9600 --slave 1</code>

Continuous level (`TE_LEVEL`) is intended for wet-well / tank control (bind to `TANK_LVL` / ALT analog). Float ladders remain available as backup.

2. Communication defaults

Setting	Factory / MooreVIEW default
Protocol	Modbus RTU (sensor is server only)
Interface	RS-485
Baud	9600
Framing	8N1 (8 data, no parity, 1 stop)
Modbus ID	1
Min gap between transactions	100 ms (<code>frameDelayMs</code>)

Recommended poll interval	500 ms (<code>pollIntervalMs</code>)
Driver I/O timeout	2000 ms

Vendor manuals: True Echo CR-L user manual (Modbus chapter — FC04 process values, FC03/16 configuration).

3. Wiring

MooreVIEW host	True Echo
RS-485 A (USB adapter or Opta A)	A / Data+
RS-485 B	B / Data–
GND	GND

Power the sensor per APG manual (typically **12–24 VDC**). Do not power the sensor from Opta logic rails.

Single-master rule: only one Modbus master on the bus (PC *or* Opta). Do not run PC USB–RS485 and Opta RS-485 master on the same A/B pair simultaneously.

4. Modbus process map (FC04 — Input Registers)

Addresses are **0-based PDU** (same as MooreVIEW `driverAddress.address`).

Address	Type	Engineering meaning	MooreVIEW tag
0	uint16	Distance (cm)	<code>TE_DIST_CM</code>
1	uint16	Distance (mm)	<code>TE_DIST_MM</code>
2	uint16	Level (cm)	<code>TE_LVL_CM</code>
3	uint16	Level (mm)	<code>TE_LVL_MM</code>
36–37	float32 CDAB	Space (selected units)	<code>TE_SPACE</code>
38–39	float32 CDAB	Level (selected units)	<code>TE_LEVEL</code>
40–41	float32 CDAB	Distance (selected units)	<code>TE_DIST</code>

4.1 Float endianness (CDAB)

APG documents float32 as **CDAB** (word-swapped big-endian):

- Register N = bytes **C D**, register $N+1$ = bytes **A B**
- IEEE bit pattern **ABCD** = `(reg[N+1] << 16) | reg[N]`

MooreVIEW `ModbusDriver` accepts `byteOrder: "CDAB"` on float32 tags. Selected units (mm / cm / m / in / ft) are configured in the APG sensor app; floats follow that unit setting.

4.2 Configuration registers (holding — reference)

Holding registers (FC03 read / FC16 write) include application type, unit setting, damping, blind zone, range, high/low level, distance offset, bus address, baud, etc. **Not mapped in v1 template** (process values only). Use the APG app or custom holding tags if site commissioning requires writes from MooreVIEW.

5. Appliance integration

5.1 Template

Field	Value
Preset id	<code>apg_true_echo_rtu</code>
Label	APG True Echo — RS485 radar level (CR-L / Plus)
Transport	<code>modbus_rtu</code>
sharedBus	false (dedicated driver id <code>apg_true_echo</code>)
Wizard group	Modbus RTU (RS-485 / USB serial)

5.2 Apply steps

1. Connect USB–RS485 to the sensor; note COM port.
2. **Drivers** → **Hardware wizard** → **Modbus RTU** → **APG True Echo**.
3. Set **serial port**, confirm baud **9600**, slave **1** (or site address).
4. **Apply** → enable driver.
5. Confirm tags update (good quality) on Tags / Trends.

5.3 Lift-station / ALT use

Recommendation	Detail
Primary analog	Bind <code>TE_LEVEL</code> → <code>TANK_LVL</code> or ALT <code>altLevelId</code>

Units	Align APG unit setting with ST thresholds (meters preferred for ALT bands)
Backup	Keep float DI ladder (LVL_* / I1 – I4) with levelInputMode digital or both
Failsafe	On Modbus BAD quality, ST should not chase stale level — use floats / hold last good per site SOP

6. Opta sample (bring-up)

Path: `firmware/arduino-opta-true-echo/OptaTrueEchoSample/`

Standalone sketch (no MQTT/ST). Polls the same FC04 map and prints to **Serial 115200**.

Define	Default	Meaning
TE_SLAVE_ID	1	Modbus address
TE_BAUD	9600	Baud
TE_POLL_MS	500	Poll period
TE_FRAME_DELAY_MS	100	Inter-frame gap

Libraries: ArduinoRS485 only.

Example Serial line:

```
[TE] dist_cm=120 dist_mm=1200 lvl_cm=80 lvl_mm=800 | space=1.200
level=0.800 dist=1.200 | ok=1 fail=0
```

Use this sketch to validate wiring and register decode before PC wizard apply.

7. Related Opta firmware (mqtt-st 2.3.81)

Production sketch `MooreviewOptaMqttSt` (syncd from `C:\data\MooreviewOptaMqttSt`) includes RS-485 **fieldbus** for **EZ Meter** (`-DMV_FIELDBUS=1 -DMV_EZMETER=1`), not True Echo process tags. True Echo on Opta for review/bring-up remains the **standalone sample** above; appliance PC master is the supported production path for radar level today.

Flag	Device
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<code>-DMV_FIELDBUS=1 -DMV_EZMETER=1</code>	EZ Meter DDS-RGB → <code>DDS_*</code> / PQ tags
True Echo sample sketch	APG radar Serial bring-up

8. Verification checklist

- ☐ Sensor powered; A/B/GND correct; no double master on bus
- ☐ `modbus-probe.js` FC04 @0x4 and @36x6 succeed; CDAB decode looks sane
- ☐ Wizard apply creates tags `TE_*` on driver `apg_true_echo`
- ☐ `TE_LEVEL` tracks known tank change (or bench target)
- ☐ Opta sample (optional) matches PC readings within expected noise
- ☐ ALT / `TANK_LVL` binding documented for the project
- ☐ Float backup verified if used

9. Code / doc references

Item	Location
Device template	<code>src/devices/templates/apg_true_echo_rtu.json</code>
Template authoring notes	<code>src/devices/templates/README.md</code> (APG section)
Float CDAB decode	<code>src/drivers/modbusDriver.js</code> (<code>_decodeFloat32</code>)
Unit tests	<code>test/apgTrueEchoTemplate.test.js</code>
Opta sample	<code>firmware/arduino-opta-true-echo/</code>
Opta mqtt-st sync	<code>scripts/sync-opta-firmware-from-data.ps1</code>
Vendor	https://apgsensors.com/

10. Review sign-off

Role	Name	Date	OK
Controls / OT			☐
Firmware			☐

Project eng.			☐
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Notes / open items:
